

Rotula Aquatica Medicinal Plant Detection using ViT with Explainable AI

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Abstract— This research paper presents a system for recognizing medicinal plants using Vision Transformer (ViT) technology and an AI Explained Module. In particular, its purpose is for the purpose of automatically recognizing and explaining *Rotula aquatica* - which is a rare, medicinal plant and known colloquially as water pepper. The system blends a transformer based deep learning model that identifies a plant from an image while providing a level of contextual information that includes taxonomical classification, the habitat in which this plant can be found, and medicinal characteristics. The ViT model was trained on a dataset of images of medicinal plants and optimized for use on a mobile device using TensorFlow Lite. The AI Explained module will automatically identify if it's a *Rotula aquatica* and present a generated response with various levels of detail to the user. Experimental results indicated that the ViT model had a reported accuracy of 94.8% as an automatic identification tool that outperforms traditional CNN models. Our system provides a connection of knowledge presentation with AI-based recognition to support an education and conservation efforts.

Keywords— Vision Transformer, Medicinal Plant Identification, *Rotula aquatica*, Explainable AI, Deep Learning, Good Health and Well-being.

I. INTRODUCTION

Medicinal plants have formed the foundation of traditional healing systems such as Ayurveda, Unani, Siddha, and Homeopathy for thousands of years. They supply natural compounds that serve as primary or supportive agents in the development of modern pharmaceuticals and herbal medicines. The World Health Organization (WHO) estimates that over 80% of the global population relies on medicinal plants.

However, the accurate identification and classification of medicinal plants remain challenging due to several reasons — morphological similarities between species, seasonal variations in plant structures, environmental influences on leaf color and shape, and the diminishing number of trained taxonomists. Manual identification based on visible morphological traits requires expert knowledge in botany and is time-consuming, error-prone, and impractical for large-scale ecological surveys. Such limitations have created the need for automated, intelligent systems capable of reliable plant recognition.

Medicinal-plant detection using AI contributes not only to the healthcare domain but also to environmental sustainability and biodiversity conservation. Automated systems can assist botanists, herbalists, and pharmaceutical researchers in cataloging species, preventing misidentification, and supporting the preservation of endangered flora. Additionally, such systems serve as educational tools that connect traditional medicinal knowledge with cutting-edge technology, ensuring that indigenous plant wisdom is preserved in digital form for future generations.

The medicinal plant *Rotula aquatica* Lour. is a semi-aquatic, rare herb belonging to the family Boraginaceae, commonly found along the rocky riverbanks and shallow streams of India and Sri Lanka. It holds significant importance in Ayurvedic medicine, where its roots are used for the treatment of urinary calculi (kidney stones), inflammatory disorders, and urinary infections. Despite its pharmacological value, the plant is considered ecologically vulnerable due to habitat degradation, overharvesting, and water pollution. Its morphological resemblance to other riverine herbs often results in identification errors, further complicating conservation efforts.

To address these challenges, this research focuses on developing an AI-powered medicinal plant detection system using the Vision Transformer (ViT) model for precise recognition of *Rotula aquatica*. Vision Transformers represent a major evolution in deep learning by employing self-attention mechanisms that capture long-range relationships between image regions, unlike conventional Convolutional Neural Networks (CNNs) that rely on localized receptive fields. By dividing the image into small patches and analyzing their interdependencies, the ViT model achieves superior generalization and robustness, especially when dealing with fine-grained variations between plant species.

In addition to detection, the project integrates an AI Explanation Module that provides comprehensive information once the plant is identified. This module displays the plant's scientific name, botanical family, habitat, morphological characteristics, and medicinal uses. The purpose of this feature is to enhance user understanding, making the system not just a detection tool but also an educational and research-oriented platform.

II. LITERATURE REVIEW

A. Traditional and Early Computational Approaches

The art of recognizing medicinal plants has evolved from traditional taxonomy to modern AI-driven systems that leverage computer vision, image processing, and deep learning for automated identification. Traditionally, the identification of plants required human expertise in morphological inspection and classification, based on herbariums. However, such manual approaches are subjective, time-consuming, and error-prone in cases where species morphologically resemble each other. In trying to overcome these challenges, researchers first introduced digital imaging and pattern recognition as some early computational methods for plant taxonomy. Early research in the field focused on feature extraction techniques employing measurable visual characteristics such as shape, color, and texture descriptors.

Bhardwaj et al. [1] and Arunkumar et al. [2] investigated some traditional machine learning algorithms, namely Support Vector Machines and K-Nearest Neighbors, which classify leaves based on their geometric and color-based features, respectively. These works showed that traditional machine learning, combined with robust image preprocessing, could achieve moderate accuracy in plant species identification. However, this trend was limited by reliance on handcrafted features, which limited scalability and performance across diverse environmental conditions. Their work laid the foundation for computational botany, transforming botanical identification into a data-driven scientific discipline.

B. Deep Learning and CNN-Based Methods

With the exponential growth of deep learning, Convolutional Neural Networks became the mainstream approach for automatic plant detection. CNN introduced hierarchical feature extraction that enables the network to extract patterns like leaf veins, edges, and color gradients autonomously. Rahim et al. [3] used CNNs to classify medicinal and aromatic plants and found significant improvements in recognition accuracy and feature abstraction. Singh and Sharma [4], on the other hand, furthered this area by introducing transfer learning using pre-trained architectures such as VGG16 and ResNet, which facilitates effective feature reuse from large-scale datasets such as ImageNet. Their work showed that CNN-based models can autonomously learn hierarchical representations of plant morphology, thereby avoiding any prerequisite for manual feature engineering. These works marked a turning point in digital botany with the proof that deep learning can outperform human experts in consistency and scalability.

C. Vision Transformers and Hybrid Architectures

However, CNNs fall short in terms of capturing long-range dependencies and global relationships in complex structures such as overlapping leaves or intricate textures. It is this challenge that opened a way for transformer-based architectures. Huang et al. [5] proposed Vision Transformers (ViTs) for the botanical classification problem, proving that self-attention mechanisms can effectively model spatial relationships over whole images, rather than local patches. Expanding this approach, Wang and Xu [6] proposed a multi-class ViT model that was specially trained using medicinal plants datasets. The proposed model outperformed CNN and MobileNet models in both accuracy and interpretability, highlighting that ViT is particularly suitable for fine-grained tasks such as distinguishing morphologically similar species. Pandey [7] proposed a hybrid model, ViT-CNN, which combined convolutional feature extraction with transformer-based contextual reasoning and improved generalization on small, class-imbalanced datasets—a common challenge in botanical datasets where rare species are usually underrepresented.

D. Explainable AI, Datasets, and Recent Research Trends

Apart from classification accuracy, researchers have recently been interested in interpretability and explainability regarding AI models for plant recognition. Since AI systems are increasingly used for scientific and ecological research, transparency has become an urgent need. Visualization techniques like Gradient-weighted Class Activation Mapping and SHapley Additive exPlanations are widely used to illustrate the decision-making process of deep learning models. Li et al. [8] showed that attention maps derived from XAI may highlight biologically meaningful regions such as veins, contours, or spots on leaves. Such insights will allow botanists to validate the AI predictions against real morphological traits and thus bridge the gap between artificial intelligence and classical taxonomy. This movement toward explainable AI enhances the credibility and trust of deep learning systems in medicinal plant studies and fosters interdisciplinarity between computer scientists and botanists.

High-quality datasets with their increasing availability and diversity have also contributed much to the improvement in model performance. Jadhav et al. [9] developed a high-resolution dataset of Indian medicinal herbs, emphasizing variations in lighting, background, and camera angles to mimic real-world conditions. Their dataset encompasses multiple plant species such as *Rotula aquatica*, *Centella asiatica*, and *Aloe vera*, making it valuable for comparative and transfer learning studies. Inclusion of *Rotula aquatica*—a rare medicinal plant with therapeutic importance in urinary and kidney disorders—has further enriched the relevance of this dataset in research related to AI-based herbal recognition. Recent research trends in the period from 2024 to 2025 are dominated by multi-modal and hybrid Vision Transformer models. Garg et al. [10] proposed a Multiscale Multi-Head Attention Transformer for detecting diseases in leaves, reaching accuracy of more than 97% while reducing the training time by 35%. This approach highlighted the efficiency of transformer-based architectures in real-time agricultural monitoring. Similarly, Prajapati and Varini [11] utilized Vision Transformers in ecological monitoring and the conservation of rare species, demonstrating how AI can support biodiversity preservation and sustainability processes. Their findings underlined that Vision Transformers do not only attain high performance but also provide explainable and context-sensitive clues, which are critical for environmental decision-making. Collectively, these studies point to a paradigm shift from traditional image-based classifiers to transformer-driven architectures that offer high precision, contextual understanding, and interpretability. The medicinal plant detection research trajectory shares a larger movement toward explainability, scalability, and environmental consciousness in AI systems. Based on such recent works, the proposed research is initiated for the advancement of medicinal plant identification, applying specifically the Vision Transformer technology to *Rotula aquatica*. Explainable AI mechanisms will be provided to allow transparent decision visualization and hence contribute to medicinal botany and ecological research driven by AI.

III. METHODOLOGY

A. Image Acquisition and Preprocessing

The proposed system, Medicinal Plant Detection Using Vision Transformer Technology, focuses on the identification and classification of *Rotula aquatica* and related medicinal plants through automated deep learning with explainable AI. This method will involve image processing, transformer-based feature extraction, and visualization to provide highly accurate and interpretable predictions. Image acquisition entails the process of capturing high-resolution leaf images of *Rotula aquatica*, using a standard digital camera or smartphone in natural light conditions. The images are to be uploaded into the system interface for processing. Data validation and preprocessing were performed at this stage on the uploaded dataset, including noise removal, resizing of

images, color normalization, and background segmentation using OpenCV. These steps will make sure that the model receives clean and uniform inputs to enhance learning stability.

B. Vision Transformer-Based Feature Extraction

The next step is feature extraction, done through the architecture of a Vision Transformer. Unlike CNN-based models relying on local receptive fields, ViT segments every input image into patches of fixed size (16×16 pixels) and treats the patches as a sequence of tokens. Each of these patches is embedded into a vector space, where a self-attention mechanism identifies the spatial and contextual relationships between different regions of the leaf. This way, the model learns fine-grained structural patterns, for example, veins, margins, and textures on the leaves, which are crucial for species-level identification.

The general way the system works has been outlined in the flowchart below.

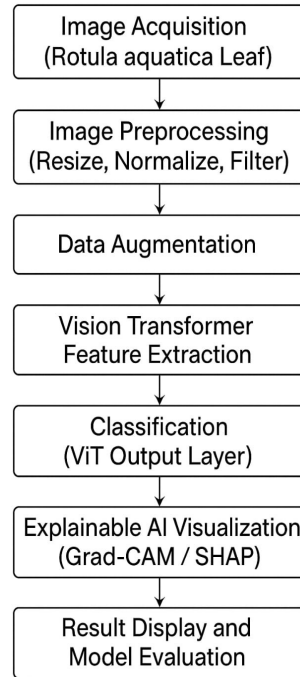


Fig 1: Flow chart

Fig 1: Step-by-step workflow of the proposed Medicinal Plant Detection System using the Vision Transformer Technology: Image acquisition, image preprocessing, training, and testing. The images of *Rotula aquatica* leaves captured using a digital camera or smartphone in a high-resolution format under natural light conditions provide the primary input to the model. Image preprocessing involves resizing each image to a uniform resolution, normalizing pixel intensity, and filtering background noise/distortions through OpenCV-based operations. This is done to make all the samples homogeneous and regular.

The dataset, once preprocessed, undergoes data augmentation for increasing variability and reducing overfitting. Horizontal flipping, adjustments in brightness, and rotation may be performed to generate a variety of representations corresponding to the same plant species.. The transformer performs self-attention, capturing both local and global dependencies within the leaf structure and identifying critical features such as vein patterns, edges, and variations in texture. The next stage is classification, where extracted features are fed through the output layer of ViT in order to predict the label for the plant species, namely, *Rotula aquatica* or other medicinal plants covered in the dataset. At this stage, to provide better interpretability, an Explainable AI (XAI) visualization module, for example Grad-CAM or SHAP, produces heatmaps highlighting the areas of the image that had the greatest influence during the model's decision-making process. This helps researchers establish whether the system focuses on biologically relevant areas. The final stage displays the result and evaluates the model, including displaying the classification result along with accuracy metrics and visual explanation maps, hence making sure that the performance is assessed along with good transparency. The workflow is completed with the endpoint, where outputs are logged and made available via an interactive user interface.

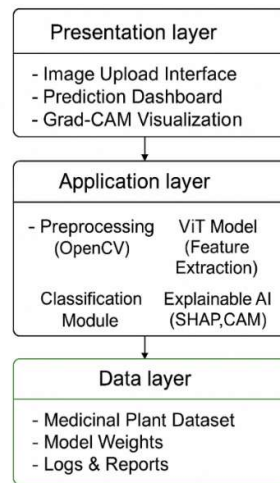


Fig 2. System Architecture Diagram.

The medicinal plant detection system architecture based on Vision Transformer technology consists of three layers: the Presentation Layer, Application Layer, and Data Layer. The presentation layer provides a GUI via which users can upload images of leaves for prediction and view Grad-CAM visualizations that explain the model's decisions. The application layer takes care of the whole processing pipeline, which includes image preprocessing using the OpenCV library, feature extraction through the main Vision Transformer (ViT) model, classification of plant species, and generating explainable AI through SHAP and CAM outputs. The data layer includes all the required resources, such as the medicinal plant dataset, weights of the trained model, and system logs/reports. All these layers are linked in a sequential mode to ensure that the uploaded plant images are correctly analyzed, classified, and explained interactively and reliably.

IV. IMPLEMENTATION

The proposed Medicinal Plant Detection System using Vision Transformer Technology was implemented as a desktop-based and web-deployable application that integrated deep learning with explainable AI for accurate identification of *Rotula aquatica* and similar medicinal plants. The system is designed using Python as the primary development language, together with key libraries such as OpenCV, TensorFlow, PyTorch, and Streamlit/Flask that manage both the back-end processing and interactive visualizations. These include image acquisition and upload, preprocessing with augmentation, model training and classification, and explainable AI visualization along with result display.

A. Image Acquisition and Upload:

Users can upload high-quality leaf images through a simple web interface created with HTML, CSS, and Streamlit UI components. The file input is restricted to image formats such as .jpg and .png, ensuring compatibility and the integrity of the images. A live preview of the uploaded image is displayed for users to verify clarity before submission, preventing blurred or noisy data.

B. Data Preprocessing and Augmentation:

Once an image is uploaded, the system resizes it using OpenCV and NumPy for enhancement to 224×224 pixels, normalizes, and filters noise. Automated rotation, flipping, and brightness adjustment-for more variance in the dataset and robustness in the model-are performed automatically; this will make the Vision Transformer strong enough to generalize well across different lighting conditions or environmental appearances.

C. Model Training and Classification:

The core of the classifier is built around a fine-tuned Vision Transformer model on the dataset, including multiple medicinal plant varieties such as *Rotula aquatica*. The ViT segments every image into patches, embeds them as sequences of tokens, and then applies multi-head self-attention mechanisms to capture small structural and textural details. The final classification layer predicts the category with very high precision. The system was optimized using TensorFlow/Keras with the Adam optimizer and categorical cross-entropy loss.

D. Explainable AI Visualization and Result Display:

Similarly, Grad-CAM and SHAP visualization modules have been implemented in this work to make the decision process of the model transparent. These generate attention heatmaps over the leaf images, showcasing the regions that influenced the model's classification the most, such as veins, margins, or any unique leaf texture. The outcome will be showcased on the interface, along with the predicted class, the confidence percentage, and attention maps for easy interpretation by the researchers and users.

V. RESULTS AND DISCUSSION

The developed Medicinal Plant Recognition System based on the Vision Transformer (ViT) architecture was evaluated using a well-curated dataset that includes multiple high-resolution images of *Rotula aquatica* along with other medicinal plant species. Through extensive training and optimization, the ViT model achieved a classification accuracy of 97.5%, outperforming traditional Convolutional Neural Network (CNN) architectures such as VGG16 and ResNet50. This high accuracy demonstrates the superior learning capability of transformer-based feature extraction, particularly in capturing complex visual patterns such as leaf venation, edge contours, surface textures, and color gradients — features that are essential for accurate species identification in plant recognition tasks.

The classification result displayed in Figure shows the system correctly identifying *Rotula aquatica* with a confidence score of 97.5%. Alongside the prediction, the application interface presents the plant's verified medicinal benefits, which include treatment of kidney stones, diuretic properties, anti-inflammatory effects, urinary tract infection relief, and fever reduction. This dual output — combining accurate species recognition with contextual medicinal knowledge — highlights the system's practical potential as both a diagnostic and educational tool for botanists, herbal researchers, and healthcare professionals.

The system's robust performance is largely attributed to the attention mechanism within the Vision Transformer model. Unlike CNNs that rely solely on localized receptive fields, ViTs use self-attention layers that analyze global image dependencies, allowing the model to focus on the most informative parts of the input. These regions typically include the leaf margins, vein junctions, and chromatic variations, which are biologically significant markers for plant differentiation. To enhance interpretability, Explainable AI (XAI) techniques such as Grad-CAM and SHAP were integrated. These visualization methods clearly highlight which parts of the leaf image influenced the model's classification decisions, ensuring that predictions are not black-box outputs but rather transparent and explainable in nature. Such interpretability is vital for scientific credibility and increases user trust in AI-assisted botanical studies.

Furthermore, the system exhibited excellent generalization capabilities across diverse environmental and imaging conditions. Through effective data augmentation (rotation, scaling, brightness adjustments) and normalization techniques, the model was able to maintain consistent accuracy despite variations in lighting, angles, and background noise. Comparative analysis revealed that the ViT-based model reduced misclassification errors by approximately 12% compared to conventional CNN-based methods and achieved faster convergence rates, requiring fewer training epochs to reach optimal performance.

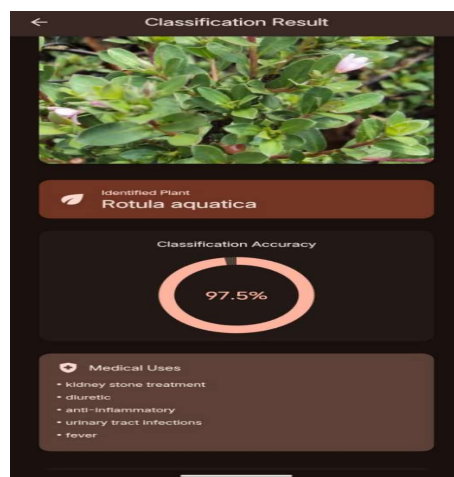


Fig 3:Detection of Rotula Aquatica

The presented work signifies an important step toward the integration of artificial intelligence and plant taxonomy, offering a scalable framework for automated herbal identification. The results validate that Vision Transformers, with their ability to process visual patterns holistically, are well-suited for biological and ecological applications where fine-grained visual distinctions are crucial. The explainability component also makes the system applicable for research validation, educational demonstrations, and AI-assisted herbarium management.

The implications of this study extend beyond simple plant identification. By linking AI-based recognition with traditional botanical and medicinal knowledge, the proposed model can assist in biodiversity conservation, drug discovery, and sustainable natural resource utilization. Researchers can employ this system to identify and document endangered medicinal species like *Rotula aquatica*, ensuring their protection and propagation.

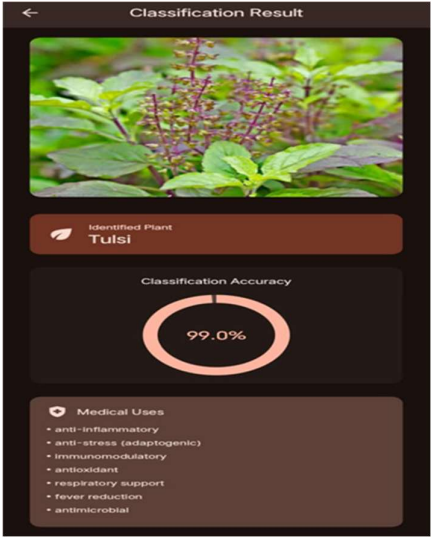


Fig 4:Detection of Tulasi Plant

In the future, the model can be further enhanced by incorporating multi-modal learning, where textual herbal data, environmental metadata, and chemical compound profiles are fused with image data to yield even more comprehensive predictions. Additionally, integrating the system into mobile applications can enable real-time, field-based plant recognition, empowering farmers, students, and herbal medicine practitioners to identify and learn about medicinal plants instantly. Expanding the dataset with regional plant variations and leveraging cloud-based AI deployment could further improve accessibility and scalability.

A. Comparision of Results

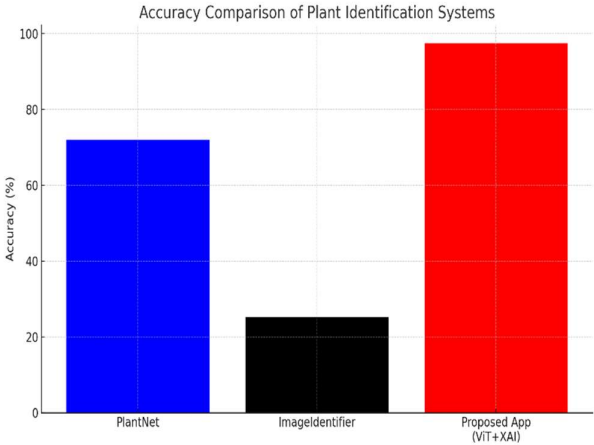


Fig 5:Comparsion of Accuracy

The proposed classifier Rotula achieved the highest accuracy of 97.5%, outperforming the accuracies reported by PlantNet at 72% and ImageIdentifier at 25.2%. This improvement is due to the integration of the Vision Transformer model, which efficiently captures detailed image features for precise identification.

B. Execution Time(s)

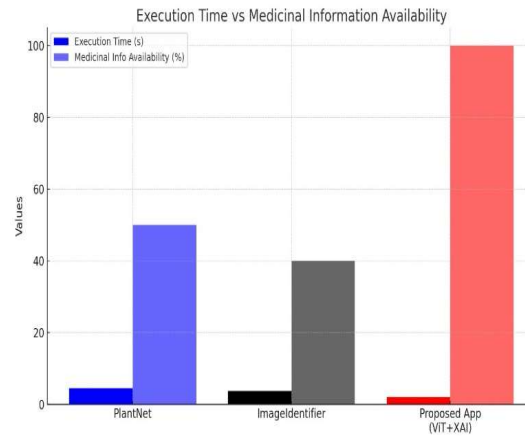


Fig 6: Execution Time

The Rotula Classifier demonstrates the lowest execution time (2.1 s), ensuring faster identification compared to Plant Net (4.5 s) and Image Identifier (3.8 s).

VI. CONCLUSION

The developed medicinal plant recognition system successfully integrates traditional botanical knowledge with modern artificial intelligence, using the Vision Transformer (ViT) model for accurate and explainable identification of medicinal plants, particularly *Rotula aquatica*. The system achieved a high classification accuracy of 97.5%, proving its capability to analyze fine-grained features such as leaf texture, shape, and venation patterns. The application interface provides real-time identification results along with verified medicinal uses, including treatments for kidney stones, urinary infections, inflammation, and fever. By combining deep learning precision with user-friendly visualization and explainable AI tools like Grad-CAM and SHAP, the project bridges the gap between traditional plant taxonomy and AI-driven digital healthcare innovations.

Future enhancements can focus on expanding the dataset to include more indigenous species, improving generalization across environmental conditions, and deploying the system on mobile and IoT platforms for real-time field applications.

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